

Monthly Intelligence Report

Edition 001 — The Uusimaa–Pirkanmaa Grid Corridor

Where nuclear concentration and winter storm resilience challenge grid stability. Inaugural edition · April 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

4,022

Major + distribution
across Fingrid

GRID LINES MAPPED

~6,665

~14,000 km
transmission &
distribution

MC SIMULATIONS

~40.2 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero
proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes ~50,540 source data points per run, executes ~40.2 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs ~151 million arithmetic operations — producing ~51,000 output data points with full uncertainty quantification across 4,022 substations and ~6,665 grid lines spanning ~15,200 km.

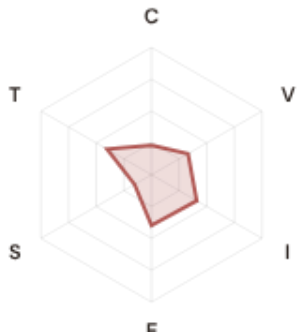
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources (Energiavirasto, Fingrid, Gasum, FMI, Tilastokeskus, Gränges, Tekes), making the methodology fully auditable and reproducible. Initially covering Finland's Fingrid-managed transmission and ~80 distribution network operators (DNOs), the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Substation_672626231

Varsinais-Suomi, Varsinais-Suomi · 110 kV

0.382 **Medium** **0.072** **37th**
R_FINAL CLASSIFICATION CI WIDTH FLEET PERCENTILE



COMPONENTS

C Continuity: 0.231 / 0.30 (77%)
I Infrastructure: 0.409 / 0.25 (164%)
S Saturation: 0.153 / 0.20 (77%)
V Voltage: 0.331 / 0.10 (331%)
E Economic: 0.399 / 0.10 (399%)
T Transition: 0.402 / 0.05 (804%)

MODIFIERS

R3 Consequence: 0.977 ↓ dampens
R4 Graph Criticality: 1.350 ↑ amplifies
R6a Restoration: 0.960 ↓ dampens
R6b Seismic: 1.023 ↑ amplifies
R7 Digital Readiness: 1.013 ↑ amplifies

This month's spotlight — automatically selected for June 2026 — is **Substation_672626231** in Varsinais-Suomi, Varsinais-Suomi. With an R_final of 0.382 (Medium band, 37th fleet percentile), its risk profile is dominated by the T (Transition) component at 804% of its weight ceiling, followed by E (Economic) at 399%. Modifiers collectively shift R_base by 4.64%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Finland's Energy Regulatory Office (Energiavirasto) and network operator strategic planning requires.

CYBER HIGH EXPOSURE

0
0.0% of fleet

BLIND SPOTS

2
High/Critical + R7 < median

AVG R7 MODIFIER

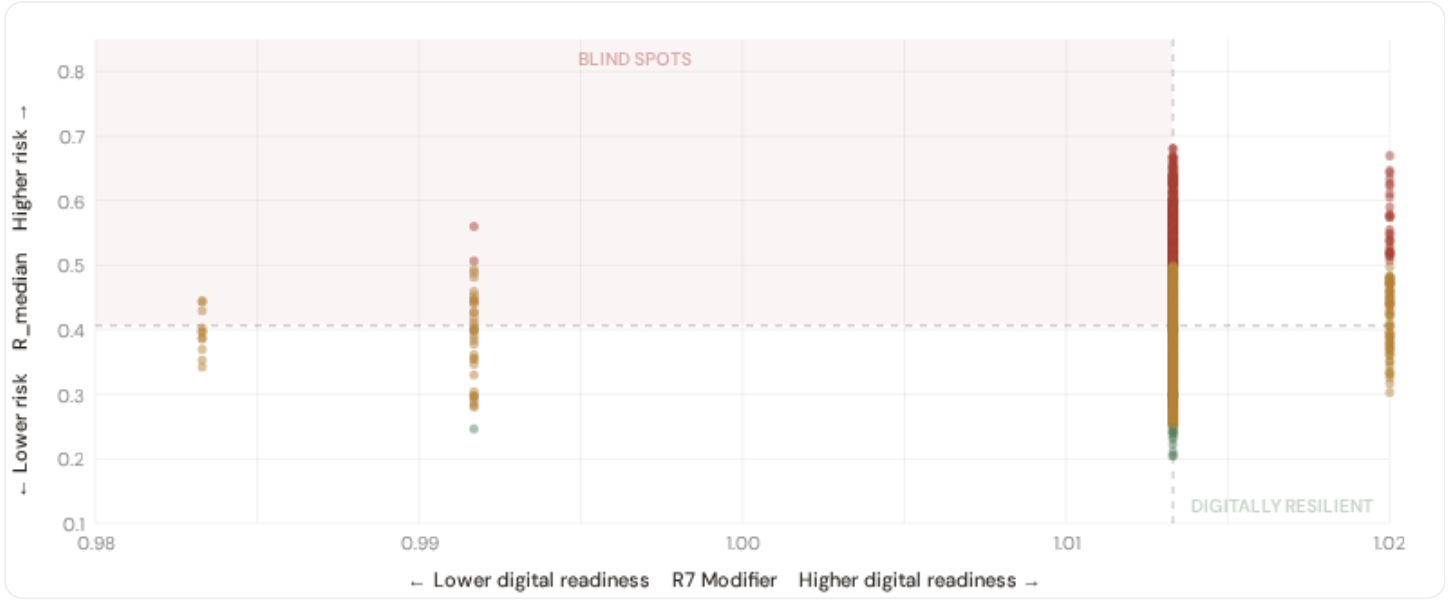
1.0132
Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

1,000
24.9% · R3 ≥ 0.98

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.

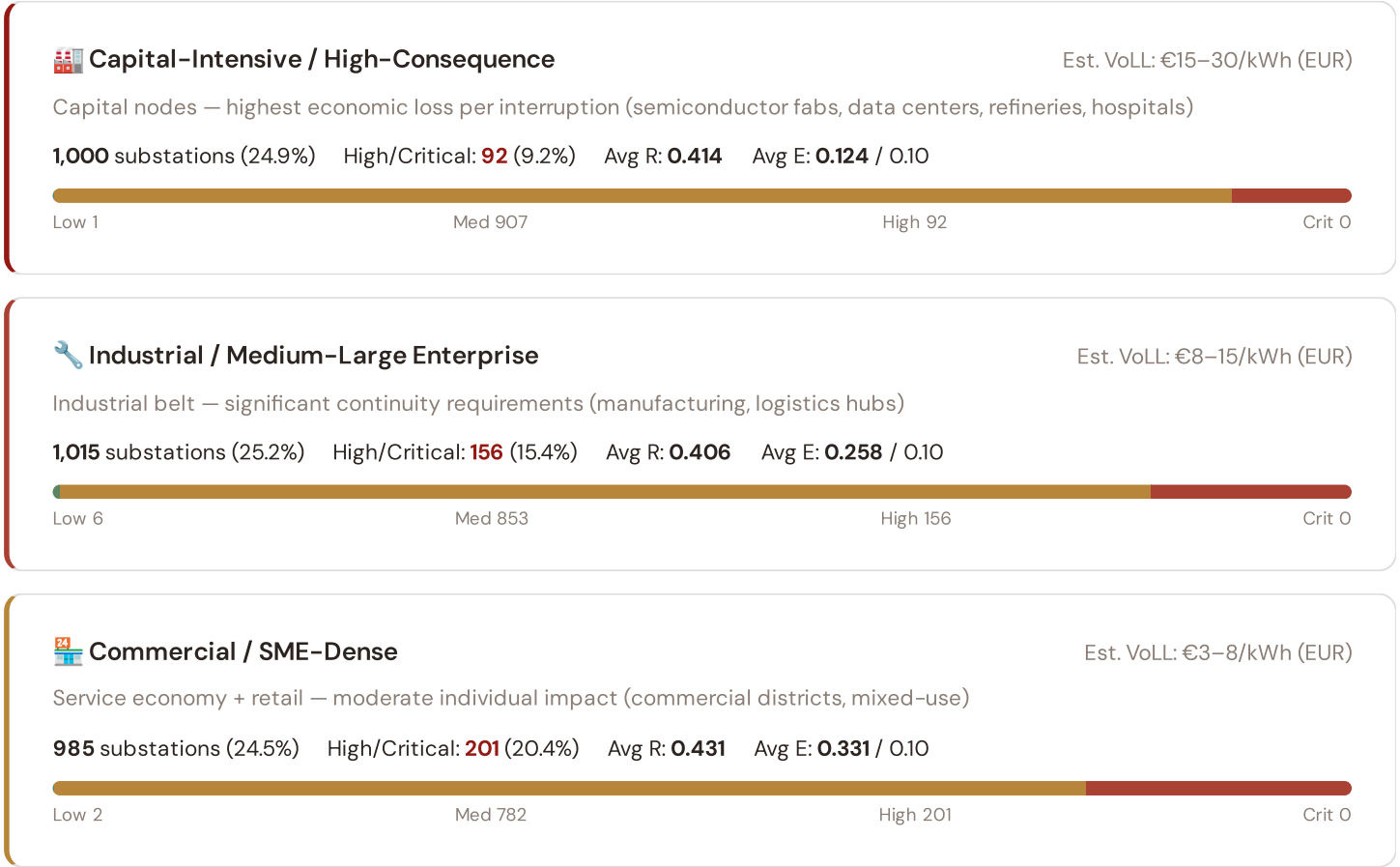


● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **2 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0133$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Lappi (1), Pohjois-Pohjanmaa (1)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Industrial Sector

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



Sparse activity, agricultural land — lower VoLL (residential, agricultural)

1,022 substations (25.4%) High/Critical: **139** (13.6%) Avg R: **0.411** Avg E: **0.365** / 0.10



Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **92 substations** combine capital-intensive economic fabric ($R3 \geq 0.98$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Uusimaa (820)**, **Varsinais-Suomi (112)**, **Pirkanmaa (68)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 5.5%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Maakunta & District Digital Readiness Ranking

Maakunnat ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_{median} to identify regions combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Åland		1.0025
2	Etelä-Pohjanmaa		1.0116
3	Lappi Δ high R		1.0121
4	Kainuu Δ high R		1.0127
5	Keski-Suomi Δ high R		1.0128
6	Pohjois-Savo Δ high R		1.0131
7	Varsinais-Suomi		1.0133
8	Päijät-Häme		1.0133
9	Pohjois-Karjala Δ high R		1.0133
10	Keski-Pohjanmaa Δ high R		1.0133
11	Kymenlaakso		1.0134
12	Pirkanmaa		1.0134
13	Uusimaa Δ high R		1.0134
14	Etelä-Karjala		1.0134
15	Kanta-Häme		1.0135

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Åland** has the lowest average R7 (1.0025), while **Satakunta** leads at 1.0138 — a spread of 0.0113 across the R7 range. **4 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

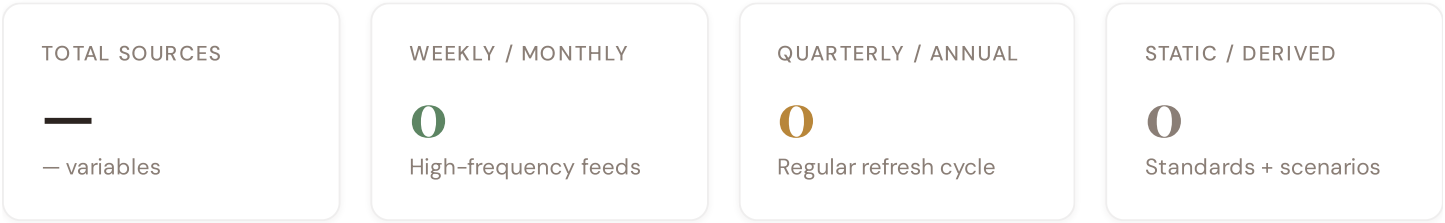
B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0132, median = 1.0133; 0 substations classified HIGH cyber-exposure; 2 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on — **verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

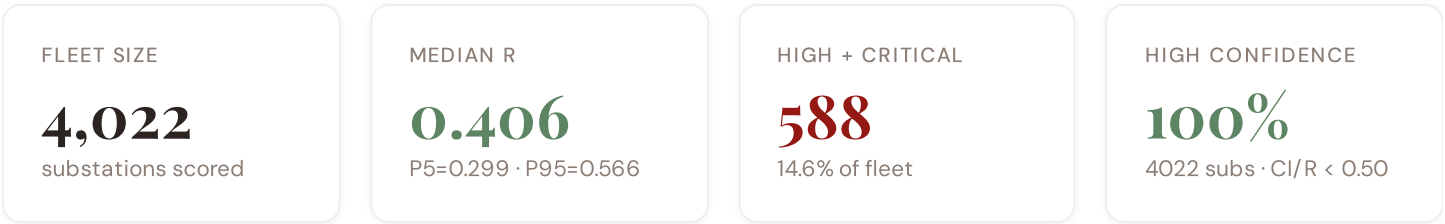


Data Source Registry — June 2026

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot



BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **4,022 substation scores remain unchanged** this edition. The fleet median R of 0.406 (mean 0.416) reflects a distribution skewed toward the lower bands, with 0.3% in Low and 85.0% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

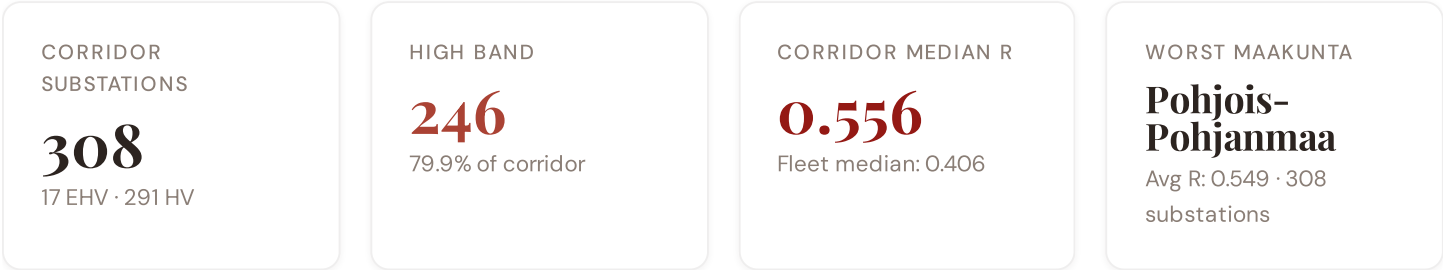
0 changes tracked across PO, PI, and P2 releases

SECTION D

The Uusimaa–Pirkanmaa Grid Corridor: Where Nuclear Concentration and Winter Storm Resilience Challenge Grid Stability

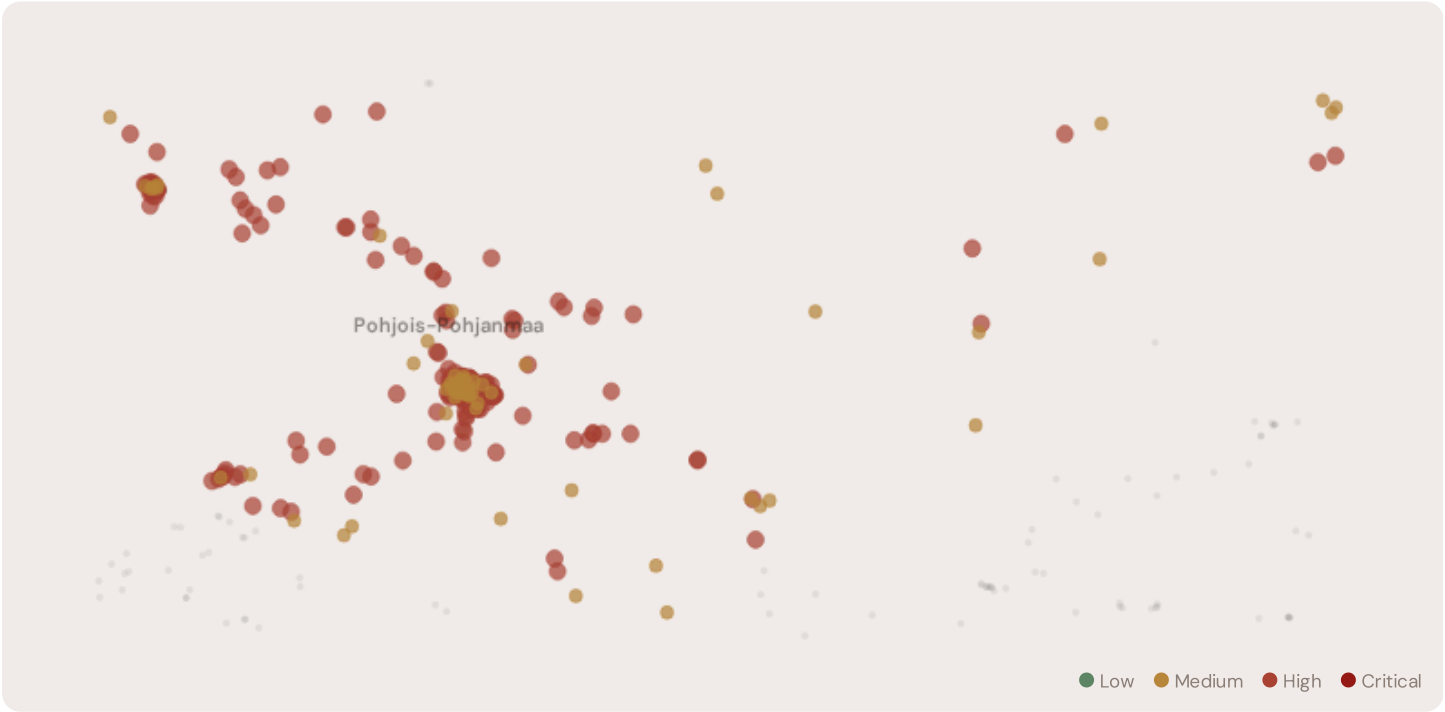
Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Uusimaa–Pirkanmaa nuclear concentration and winter storm resilience corridor · **002** Eastern Finland DER & EU funding mismatch · **003** Mining subsidence infrastructure stress · **004** Metropolitan–Rural energy poverty disparity · **005** Grid stranding risk (nuclear-dependent assets) · **006** DSO smart meter rollout vs grid stress · **007** Gulf of Finland coastal flooding resilience · **008** T-component forward projections · **009** Maakunta Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Uusimaa–Pirkanmaa Corridor, Why Now



The Pohjois–Pohjanmaa hosts **308 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **246 substations (79.9%)** are classified High, with only **0** in the Low band. 98% of corridor substations score above the national fleet median of 0.406. The corridor median R of 0.556 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



Substation	District	R_Final	Band	C	V	I	E	S	T	Alert
Karihaara sähköasema	Pohjois-Pohjanmaa	0.681	High	0.650	0.381	0.645	0.299	0.253	0.389	
Simojoki sähköasema	Pohjois-Pohjanmaa	0.670	High	0.621	0.252	0.734	0.260	0.232	0.507	I
Ylä-Siirtolan sähköasema	Pohjois-Pohjanmaa	0.668	High	0.511	0.462	0.685	0.429	0.344	0.419	I
Sorosenperä sähköasema	Pohjois-Pohjanmaa	0.667	High	0.424	0.259	0.750	0.429	0.388	0.506	I
Utasen sähköasema	Pohjois-Pohjanmaa	0.666	High	0.521	0.220	0.750	0.415	0.357	0.448	I
Substation_910461387	Pohjois-Pohjanmaa	0.662	High	0.554	0.264	0.746	0.337	0.276	0.445	I
11 Kansakoulu	Pohjois-Pohjanmaa	0.658	High	0.613	0.159	0.740	0.300	0.304	0.370	I
Substation_1155600096	Pohjois-Pohjanmaa	0.653	High	0.371	0.384	0.750	0.348	0.398	0.513	I
Substation_1156160212	Pohjois-Pohjanmaa	0.653	High	0.402	0.323	0.746	0.409	0.411	0.429	I
M30 Ylitalo	Pohjois-Pohjanmaa	0.652	High	0.442	0.215	0.736	0.306	0.422	0.499	I
... 298 additional substations — explore full corridor in Digital Twin →										

D.3 Component Analysis

Component Averages — Pohjois-Pohjanmaa vs Fleet

C — Continuity	0.422 vs 0.315 (+34%)
I — Infrastructure	0.645 vs 0.421 (+53%)
S — Saturation	0.251 vs 0.204 (+23%)
V — Voltage	0.250 vs 0.366 (−32%)
E — Economic	0.300 vs 0.270 (+11%)
T — Transition	0.436 vs 0.400 (+9%)

Modifier Averages — Pohjois-Pohjanmaa vs Fleet

R3 Consequence	0.948	fleet 0.957 ↓
R4 Graph Crit.	1.331	fleet 1.341 ↑
R6a Restoration	1.007	fleet 0.972 →
R6b Seismic	1.029	fleet 1.018 ↑
R7 Digital Read.	1.013	fleet 1.013 ↑

The dominant risk driver across the Pohjois-Pohjanmaa corridor is **T (Transition)**, averaging 0.436 against a fleet average of 0.400 — occupying 871% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.300 vs fleet 0.270 (300% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.948 (fleet: 0.957), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.029, capturing the local seismic exposure — a dimension invisible to every competing framework.

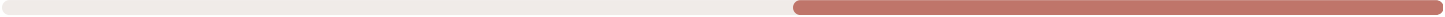
D.4 Maakunta Risk Profile

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Pohjois-Pohjanmaa

308 subs · avg R = 0.549 · H:246 M:62



The region regions-level breakdown reveals significant intra-regional variation. **Pohjois-Pohjanmaa** leads with a mean R of 0.549 across 308 substations, while **Pohjois-Pohjanmaa** scores 0.549 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Pohjois-Pohjanmaa is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications for Finland's Energy Transition

10 Pohjois-Pohjanmaa substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Pohjois-Pohjanmaa sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

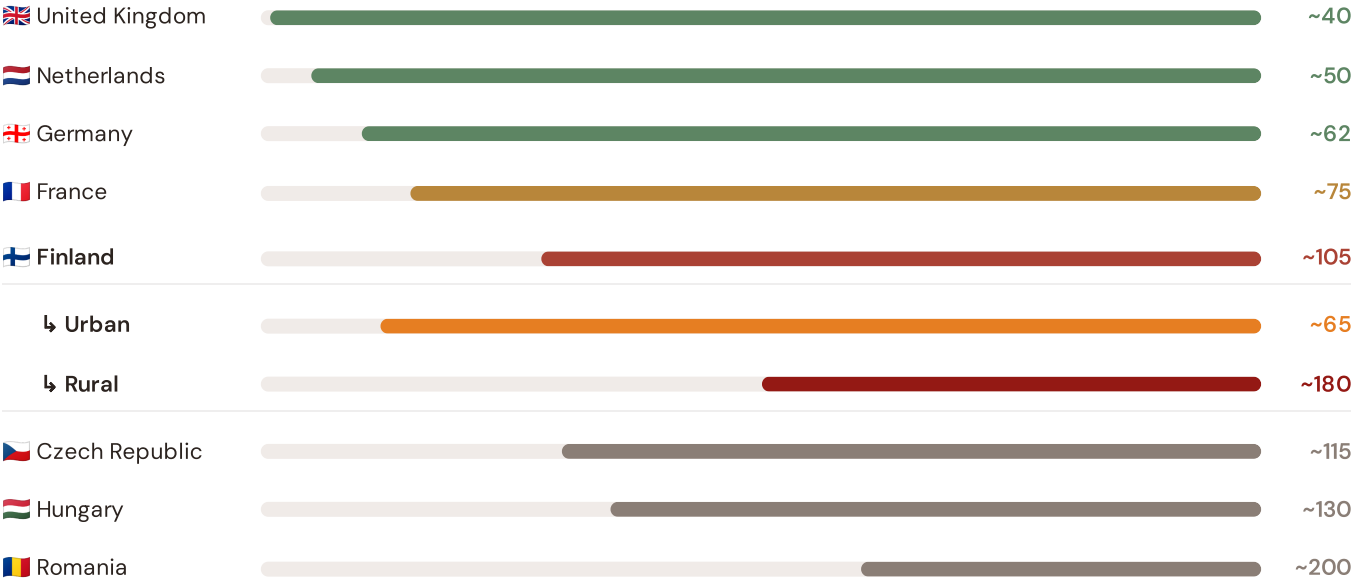
SECTION E — RECURRING MODULE

European Context Card

How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because Finland's nuclear concentration and winter storm resilience and grid resilience depend on understanding European performance standards. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Central Europe peers (003), CMIP6 climate hazard vs Nordic models (004), etc. The underlying data updates annually with Energiavirasto and national regulator publications.

Unplanned SAIDI (min/customer/year) — Selected European Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



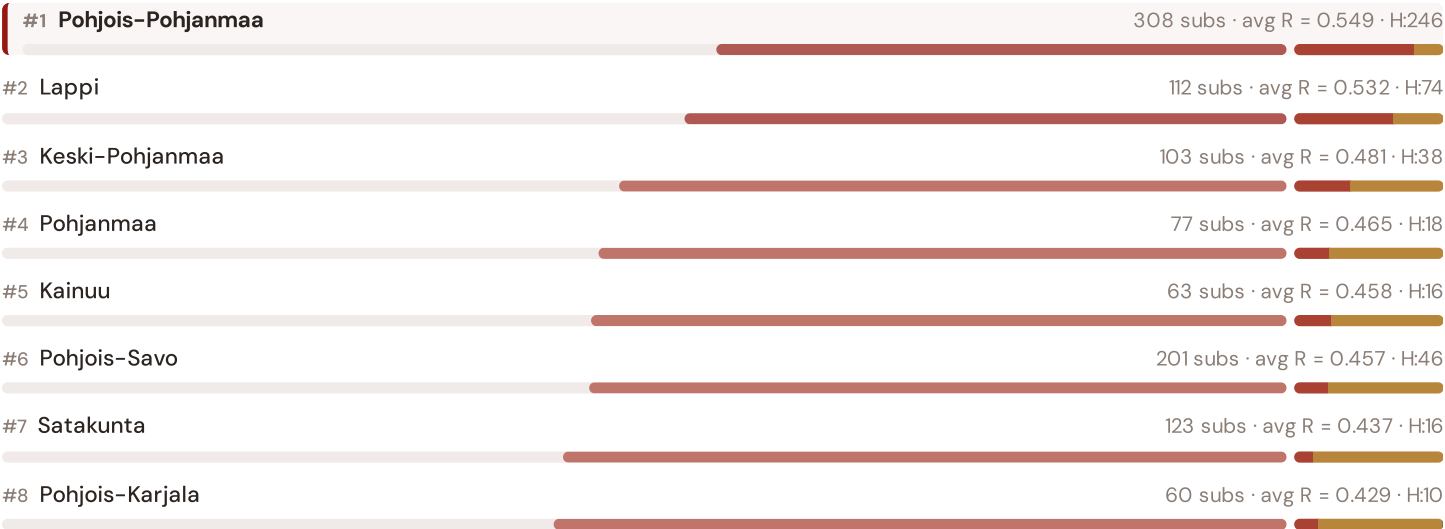
Source: CEER Benchmarking Report on Quality of Electricity Supply (2024); EEnergyvirastoLECTRIC SAIDI reports; national regulator publications. Finland sub-categories from Energiavirasto urban/rural classifications. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually.

This month's key insight: The Pohjois–Pohjanmaa corridor deep-dive shows **305 substations** (of 308) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.422 is **1.3× the fleet average** of 0.315. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Maakunta Risk Ranking — All 19 Finnish Maakunnat

Where does the Uusimaa–Pirkanmaa corridor sit within the national landscape? Maakunnat sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



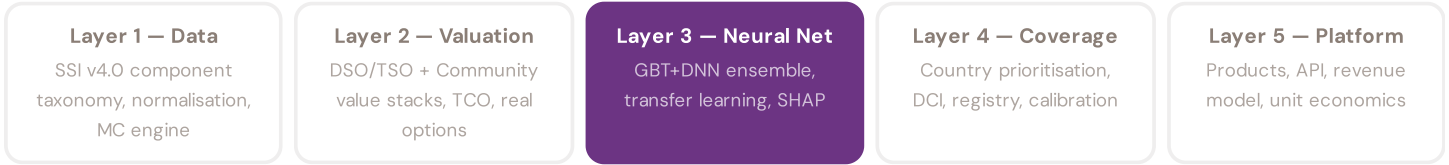


The Pohjois-Pohjanmaa ranks **#1 of 19 region regions** by mean R_{median} , placing it among the upper tier of national risk. The three most stressed are Pohjois-Pohjanmaa, Lappi, Keski-Pohjanmaa, while the three most resilient are Åland, Etelä-Karjala, Kymenlaakso. 9 of 19 score above the fleet average of 0.416. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to renewable energy investment valuation. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a battery energy storage or renewable retrofit worth at this substation — to the DSO/Fingrid and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI

Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- SHAP: game-theoretic feature attribution per prediction
- Waterfall: fleet average → substation prediction
- Direct traceability to SSI components and data sources
- Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 4,022 substations with complete feature vectors. Average CI_width of 0.069 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 14 Low, 3420 Medium, 588 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — o Quarterly Sources

none are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **TBD** (? variables →).

FLEET SPOTLIGHT

Pohjois-Pohjanmaa: Highest Risk Concentration

Pohjois-Pohjanmaa has the highest concentration of High/Critical substations: **246 of 308 (80%)** are in the upper risk bands — avg R = 0.549. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.